

TARNOW, Poland

WMO: 125750

Lat: 50.0299N

Lon: 20.9839E

Elev: 686

StdP: 14.34

Time Zone: 1.00 (EUC)

Period: 04-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) 4.1	(c) 11.4	(d) -0.8	(e) 5.4	(f) 4.9	(g) 5.7	(h) 7.6	(i) 13.9	(j) 12.6	(k) 35.0	(l) 11.7	(m) 33.9	(n) 2.0	(o) 80	(p) 0.459

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	20.0	87.9	69.4	84.5	67.8	81.5	66.2	71.2	84.1	69.6	81.3	68.0	78.4	4.7	150

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
67.1	102.2	75.9	65.6	96.9	74.1	64.1	91.9	73.0	35.4	84.3	34.0	81.6	32.7	78.5	77.9

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a) 11.2	(b) 9.8	(c) 8.8	(e) -1.3	(f) 93.5	(g) 9.1	(h) 3.1	(i) -7.8	(j) 95.8	(k) -13.1	(l) 97.6	(m) -18.2	(n) 99.4	(o) -24.8	(p) 101.6		
(4) 11.2	(5) 9.8	(6) 8.8	(7) -1.8	(8) 74.3	(9) 9.1	(10) 1.6	(11) -8.3	(12) 75.5	(13) -13.6	(14) 76.4	(15) -18.7	(16) 77.3	(17) -25.2	(18) 78.5		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	49.4	29.6	31.8	39.4	50.3	57.7	64.4	67.6	66.6	58.4	49.9	41.9	34.3		
	DBStd	15.32	10.20	9.94	9.21	7.13	7.11	6.57	5.97	5.97	6.40	7.54	7.97	8.25		
	HDD50	2437	633	509	342	81	15	0	0	0	8	99	262	488		
	HDD65	6020	1098	928	794	445	245	89	43	52	212	468	694	952		
	CDD50	2220	0	1	13	89	253	433	544	513	260	96	17	1		
	CDD65	332	0	0	0	2	18	73	123	101	15	0	0	0		
	CDH74	3380	0	0	0	51	249	728	1202	979	166	5	0	0		
	CDH80	987	0	0	0	9	45	199	386	324	24	0	0	0		
	WSAvg	4.0	4.3	4.4	4.7	4.3	4.1	3.9	3.6	3.3	3.4	3.7	4.2	4.4		
	PrecAvg	28.3	1.4	1.2	1.5	2.0	3.2	4.1	4.2	3.5	2.2	1.7	1.7	1.6		
(15) Precipitation	PrecMax	40.3	3.6	3.5	3.2	4.7	6.5	8.1	11.6	6.8	4.1	7.5	3.3	3.0		
	PrecMin	18.2	0.3	0.0	0.1	0.5	1.4	1.6	1.3	0.8	0.3	0.1	0.6	0.4		
	PrecStd	5.6	0.8	0.7	0.7	1.1	1.3	1.7	2.3	1.5	1.0	1.4	0.6	0.7		
	PrecStd	5.6	0.8	0.7	0.7	1.1	1.3	1.7	2.3	1.5	1.0	1.4	0.6	0.7		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	52.2	57.3	67.3	80.3	84.6	90.6	92.7	91.8	82.8	74.5	63.2	54.1		
		MCWB	46.3	46.7	51.9	59.6	64.7	71.0	71.1	70.0	65.6	60.4	54.2	47.4		
	2%	DB	48.2	51.2	61.4	73.3	80.4	85.4	87.8	87.3	78.9	70.6	59.5	50.9		
		MCWB	44.1	45.0	49.3	56.1	63.0	68.5	69.6	69.3	64.3	58.8	52.3	45.6		
	5%	DB	45.0	48.0	57.1	69.0	76.3	81.4	84.2	83.5	74.7	66.7	56.5	48.1		
		MCWB	41.0	42.4	47.1	54.2	62.1	66.2	67.7	67.9	62.0	57.3	50.7	43.7		
	10%	DB	42.1	44.7	53.2	64.7	72.0	78.0	80.9	79.6	71.0	62.3	53.5	45.2		
		MCWB	38.8	40.3	45.3	52.1	59.6	65.1	66.3	66.4	60.6	55.3	48.7	41.5		
	0.4%	WB	47.4	48.4	53.0	60.6	67.3	72.9	73.1	73.2	67.9	62.4	55.3	48.6		
		MCDB	51.0	53.9	63.2	77.0	80.1	86.5	87.2	86.1	80.1	71.9	60.8	53.4		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	2%	WB	44.4	45.2	50.4	57.6	64.6	70.3	71.2	70.7	65.4	59.7	53.4	46.0		
		MCDB	47.7	50.6	59.9	70.2	76.9	82.3	84.2	83.6	75.3	68.2	58.6	50.2		
	5%	WB	41.4	42.9	48.0	55.4	62.7	68.3	69.7	69.0	63.4	58.0	51.2	44.1		
		MCDB	44.4	47.3	55.4	66.5	74.3	78.2	80.9	81.0	72.1	65.2	55.8	47.7		
	10%	WB	39.2	40.8	45.8	53.5	60.6	66.2	68.0	67.4	61.6	56.0	48.9	41.8		
		MCDB	41.8	44.4	52.1	63.1	70.5	75.4	77.9	77.5	69.0	61.9	52.8	45.0		
(35) Mean Daily Temperature Range	5% DB	MDBR	9.9	11.8	15.4	19.9	19.5	19.5	20.0	20.2	18.9	16.3	11.5	9.6		
		MCDBR	11.7	16.9	22.7	28.3	26.6	25.4	26.0	26.3	25.8	23.9	16.2	12.2		
		MCWBR	8.7	10.9	12.4	13.6	12.0	10.7	10.3	10.7	12.6	13.3	10.1	8.7		
	5% WB	MCDBR	11.0	14.8	20.5	25.3	24.9	22.5	23.0	23.8	22.0	21.7	14.4	11.9		
		MCWBR	8.7	10.2	11.8	12.7	12.0	10.2	9.9	10.2	11.4	12.7	9.4	8.7		
			(d) 9.9	(e) 11.8	(f) 15.4	(g) 19.9	(h) 19.5	(i) 19.5	(j) 20.0	(k) 20.2	(l) 18.9	(m) 16.3	(n) 11.5	(o) 9.6		
(40) Clear-Sky Solar Irradiance	taub		0.316	0.340	0.384	0.438	0.430	0.432	0.450	0.444	0.410	0.373	0.346	0.311		
	taud		2.278	2.242	2.212	2.116	2.193	2.224	2.186	2.203	2.259	2.338	2.348	2.335		
	Ebn at Noon		224	248	257	255	263	263	256	250	246	234	212	209		
	Edh at Noon		23	30	37	45	44	43	44	41	35	27	21	19		
(44) All-Sky Solar Radiation	RadAvg		272	514	876	1285	1575	1679	1605	1468	1039	626	306	207		
	RadStd		43	82	112	148	192	144	191	113	150	89	37	27		

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (34 neighbors)		+0.97	N/A	N/A	+1.62	+0.83	+0.72	-227	-306	+135	+54

Nomenclature: See separate page